

$$Q1) a) \int_0^{\infty} (n^2 + 4) e^{-2n^2} \cdot dn$$

$$\text{Formula} \Rightarrow \int_0^{\infty} e^{-x} x^{t-1} dt = \Gamma t \quad \int_0^{\infty} e^{-x} x^t dt = \Gamma(t+1)$$

$$\int_0^{\infty} e^{-2n^2} n^2 dn + \int_0^{\infty} 4e^{-2n^2} dn$$

$$2n^2 = t \quad dn = \frac{1}{2\sqrt{2t}}$$

$$n = \sqrt{t/2}$$

x	t
0	0
0	∞

$$- \int_0^{\infty} e^{-t} \frac{t}{2} \frac{1}{2\sqrt{2t}} dt + - \int_0^{\infty} 4e^{-t} \frac{1}{2\sqrt{2t}} dt.$$

$$\frac{2}{\Gamma_2 \times \Gamma_1} = \underline{\underline{\Gamma_2}}$$

$$- \frac{1}{4\sqrt{2}} \int_0^{\infty} e^{-t} t^{1-1/2} dt - \Gamma_2 \int_0^{\infty} e^{-t} t^{-1/2} dt.$$

$$- \frac{1}{4\sqrt{2}} \int_0^{\infty} e^{-t} t^{1/2} dt - \Gamma_2 \int_0^{\infty} e^{-t} t^{1-3/2} dt.$$

$$1-x = -1/2$$

$$-x = -1/2 - 1$$

$$x = 3/2$$

$$= - \frac{1}{4\sqrt{2}} \int_0^{\infty} e^{-t} t^{1-1/2} dt - \Gamma_2 \int_0^{\infty} e^{-t} t^{1-3/2}$$

$$1-x = 1/2$$

$$-x = -1/2$$

$$x = 1/2$$

$$= - \frac{1}{4\sqrt{2}} \Gamma(1/2) - \Gamma_2 \Gamma(3/2)$$

$$= - \frac{\sqrt{\pi}}{4\sqrt{2}} - \Gamma_2 \frac{1}{2} \Gamma(1/2)$$

$$= \frac{-\frac{\sqrt{\pi}}{4\sqrt{2}} - \frac{\sqrt{2}}{2} \frac{\sqrt{\pi}}{2}}{\underline{\underline{\quad}}}$$

⑤ $\int_0^1 \frac{dn}{\sqrt{-\log n}}$

let $t = -\log n$

$n = e^{-t}$

$dn = -e^{-t} dt$

$\sqrt{-\log n} = \sqrt{t}$

n	t
0	∞
1	0

$$I = \int_0^1 \frac{-e^{-t} dt}{\sqrt{t}} \Rightarrow \int_0^{\infty} \frac{e^{-t} dt}{\sqrt{t}}$$

$$\int_0^{\infty} e^{-t} (t)^{-1/2} dt$$

$$\int_0^{\infty} e^{-t} t^{-1/2} = \Gamma^{1/2} = \underline{\underline{\sqrt{\pi}}}$$